Decision and Social Choice

PE3101P — Lecture 12: Revision

Instructor: Tomi Francis

National University of Singapore

5th November, 2026

Test 2

- Thursday 12 November, 14:00, in this room (AS1-0203).
- Lectures 7–11.
- Same format as Test 1: multiple-choice questions, then three of four longer questions.
- A practice paper (the longer questions) is on the course page.
- Assignment 2 is due Tuesday 10 November, 23:59.

Today

1. Internal Consistency Conditions
2. Voting Theory
3. Proof of Arrow's Theorem
4. The Liberal Paradox

Choice Functions

- A menu (or option set) S: the options available.
- $C(S)$: what may be chosen from S. It is a non-empty subset of S.
- A choice function C gives a $C(S)$ for every menu S.

Contraction Consistency (Property α)

Contraction consistency (Property α)

If x is chosen from a menu S, and x is on a smaller menu T inside S, then x is chosen from T.

$$[x \in C(S) \text{ and } x \in T \subset S] \Rightarrow x \in C(T)$$

- Taking other options off the menu can't stop x being chosen.

Contraction Consistency: Dessert

- A diner is offered apple pie (a) or blueberry pie (b).[1] He orders apple.
- The waiter comes back: there is also cherry pie (c). "In that case, I'll have the blueberry."

Menu	He chooses

[1] Attributed to Sidney Morgenbesser.

Contraction Consistency: Dessert

- A diner is offered apple pie (a) or blueberry pie (b).[1] He orders apple.
- The waiter comes back: there is also cherry pie (c). "In that case, I'll have the blueberry."

Menu	He chooses
$\{a, b\}$	a
$\{a, b, c\}$	b

[1] Attributed to Sidney Morgenbesser.

Contraction Consistency: Dessert

- A diner is offered apple pie (a) or blueberry pie (b).[1] He orders apple.
- The waiter comes back: there is also cherry pie (c). "In that case, I'll have the blueberry."

Menu	He chooses
$\{a, b\}$	a
$\{a, b, c\}$	b

b is chosen from $\{a, b, c\}$, and b is on the smaller menu $\{a, b\}$, but it isn't chosen there: contraction consistency fails.

[1] Attributed to Sidney Morgenbesser.

Contraction Consistency: The Last Apple

- A guest at dinner sees one apple left in the fruit basket. She could take it (y) or take nothing (x). To be polite, she takes nothing.[1]
- If there had been two apples (y and z), she could take one without being rude.

Menu	She chooses

[1] Sen (1993: 501).

Contraction Consistency: The Last Apple

- A guest at dinner sees one apple left in the fruit basket. She could take it (y) or take nothing (x). To be polite, she takes nothing.[1]
- If there had been two apples (y and z), she could take one without being rude.

Menu	She chooses
$\{x, y\}$	x, nothing
$\{x, y, z\}$	y or z, an apple

[1] Sen (1993: 501).

Contraction Consistency: The Last Apple

- A guest at dinner sees one apple left in the fruit basket. She could take it (y) or take nothing (x). To be polite, she takes nothing.[1]
- If there had been two apples (y and z), she could take one without being rude.

Menu	She chooses
$\{x, y\}$	x, nothing
$\{x, y, z\}$	y or z, an apple

y is chosen from $\{x, y, z\}$, and y is on $\{x, y\}$, but it isn't chosen there: contraction consistency fails.
But nothing here seems irrational: she is being polite.

[1] Sen (1993: 501).

Expansion Consistency (Property γ)

Expansion consistency (Property γ)

If x is chosen from a menu S_1, and x is chosen from a menu S_2, then x is chosen from the menu that combines them.

$$[x \in C(S_1) \text{ and } x \in C(S_2)] \Rightarrow x \in C(S_1 \cup S_2)$$

- If x survives against everything on S_1, and against everything on S_2, it survives against them all together.

Expansion Consistency: The Last Apple Again

Menu	She chooses
$\{x, y\}$	x, nothing

With one apple, y, the guest takes nothing.

Expansion Consistency: The Last Apple Again

Menu	She chooses
$\{x, y\}$	x, nothing
$\{x, z\}$	x, nothing

With the other apple alone, z, she takes nothing too.

Expansion Consistency: The Last Apple Again

Menu	She chooses
$\{x, y\}$	x, nothing
$\{x, z\}$	x, nothing
$\{x, y, z\}$	y or z, an apple

With both apples, she takes one.

Expansion Consistency: The Last Apple Again

Menu	She chooses
$\{x, y\}$	x, nothing
$\{x, z\}$	x, nothing
$\{x, y, z\}$	y or z, an apple

x is chosen from $\{x, y\}$ and from $\{x, z\}$,
but not from $\{x, y\} \cup \{x, z\} = \{x, y, z\}$: expansion consistency fails too.

Property β

Property β

If x and y are both chosen from T, and $T \subset S$, then y is chosen from S if and only if x is.

$$[x, y \in C(T) \text{ and } T \subset S] \Rightarrow [y \in C(S) \Leftrightarrow x \in C(S)]$$

- If x and y are both chosen from a smaller menu, then on a bigger menu they stand or fall together.

Property β: A Trip

- A trip to New York (x) or to Sydney (y): each good in its own way. Either would be fine.
- Then a third option: Sydney, plus a free dollar (y^+).

Menu	May be chosen

Property β: A Trip

- A trip to New York (x) or to Sydney (y): each good in its own way. Either would be fine.
- Then a third option: Sydney, plus a free dollar (y^+).

Menu	May be chosen
$\{x, y\}$	x or y

From $\{x, y\}$, either trip is fine.

Property β: A Trip

- A trip to New York (x) or to Sydney (y): each good in its own way. Either would be fine.
- Then a third option: Sydney, plus a free dollar (y^+).

Menu	May be chosen
$\{x, y\}$	x or y
$\{x, y, y^+\}$	x or y^+

From $\{x, y, y^+\}$, y shouldn't be chosen: y^+ is better.
x is still fine.

Property β: A Trip

- A trip to New York (x) or to Sydney (y): each good in its own way. Either would be fine.
- Then a third option: Sydney, plus a free dollar (y^+).

Menu	May be chosen
$\{x, y\}$	x or y
$\{x, y, y^+\}$	x or y^+

x and y are both chosen from $\{x, y\}$, but from $\{x, y, y^+\}$ only x of the two is:
Property β fails.

From a Choice Function to a Social Preference Relation

The social preference relation $\succsim_C$

$x \succsim_C y$ if and only if $x \in C(\{x, y\})$.

- So $x \succ_C y$ when x alone is chosen from $\{x, y\}$, and $x \sim_C y$ when both are.
- $\succsim_C$ is complete, by construction: something is always chosen from $\{x, y\}$, so $x \succsim_C y$ or $y \succsim_C x$.
- Every choice function gives a social preference relation in this way.

Binary-Representable Choice Functions

Binary representability

C is *binary-representable* when there is a social preference relation $\succsim$ such that, from every menu S, $C(S)$ is the set of options on S that nothing on S is strictly better than. Then $\succsim$ *represents* C.[1]

- Every choice function determines a social preference relation, $\succsim_C$.
- But not every choice function is determined by a social preference relation: not every choice function is binary-representable.

[1] Also "normal" or "binary" (Sen 1971: 309; 1993: 500).

Binary Representability: Examples

- If a complete social preference relation represents C, it is $\succsim_C$: what it says about x against y has to match what C chooses from $\{x, y\}$.
- The dessert choices aren't binary-representable: $\succsim_C$ has $a \succ_C b$, so it would never choose b from $\{a, b, c\}$.
- The trip choices are represented by $\succsim_C$, which counts x and y as equally good, since both are chosen from $\{x, y\}$.

From α and β to γ

From α and β to γ

If C satisfies contraction consistency (Property α) and Property β, it satisfies expansion consistency (Property γ).

- Suppose $x \in C(S_1)$ and $x \in C(S_2)$. We show $x \in C(S_1 \cup S_2)$.
- $C(S_1 \cup S_2)$ is not empty, so it contains some option y. If y is x, we are done.
- Otherwise, y is in S_1 or in S_2: say S_1. y is chosen from $S_1 \cup S_2$, and is in S_1, which is inside $S_1 \cup S_2$; so $y \in C(S_1)$ (Property α).
- So x and y are both chosen from S_1, which is inside $S_1 \cup S_2$; and y is chosen from $S_1 \cup S_2$. So x is chosen from $S_1 \cup S_2$ too (Property β). $\square$

From α and γ to Binary Representability

From α and γ to binary representability

If C satisfies contraction consistency (Property α) and expansion consistency (Property γ), then $\succsim_C$ represents C.[1]

- Take any menu S and any x on S. We show: $x \in C(S)$ if and only if nothing on S is strictly better than x by $\succsim_C$.
- Suppose $x \in C(S)$, and take any y on S. $\{x, y\}$ is inside S, so $x \in C(\{x, y\})$ (Property α): $x \succsim_C y$. So nothing on S is strictly better than x.

[1] Sen (1971: 314).

From α and γ to Binary Representability (2)

- Now suppose nothing on S is strictly better than x by $\succsim_C$, and take any y on S.
- If x were not in $C(\{x, y\})$, then $C(\{x, y\})$ would be $\{y\}$, since it is not empty; and then $y \succ_C x$, which is false. So $x \in C(\{x, y\})$.
- Combining the menus $\{x, y\}$, for every y on S, gives S. So $x \in C(S)$ (Property γ, applied again and again). $\square$

Binary Representability Gives α and γ

The other way round

If a social preference relation represents C, then C satisfies contraction consistency (Property α) and expansion consistency (Property γ).

- Property α: suppose $x \in C(S)$ and x is on T, inside S. Nothing on S is strictly better than x, so nothing on T is either. So $x \in C(T)$.
- Property γ: suppose $x \in C(S_1)$ and $x \in C(S_2)$. Nothing on S_1 is strictly better than x, and nothing on S_2 is, so nothing on $S_1 \cup S_2$ is. So $x \in C(S_1 \cup S_2)$. □

From α and β to Binary Representability

Corollary

If C satisfies contraction consistency (Property α) and Property β, then $\succsim_C$ represents C.

- α and β give γ. Then α and γ give binary representability. $\square$

From α and β to Transitivity

From α and β to transitivity

If C satisfies contraction consistency (Property α) and Property β, then $\succsim_C$ is transitive.[1]

- Suppose $x \succsim_C y$ and $y \succsim_C z$, with x, y and z all different. (If two are the same, $x \succsim_C z$ is immediate.)
- We show $x \in C(\{x, y, z\})$. Then $x \in C(\{x, z\})$ (Property α): $x \succsim_C z$.

[1] Sen (1971: 310, 314).

From α and β to Transitivity (2)

- If y is chosen from $\{x, y, z\}$: then $y \in C(\{x, y\})$ (Property α), and $x \in C(\{x, y\})$, since $x \succsim_C y$. So x is chosen from $\{x, y, z\}$ too (Property β).
- If z is chosen from $\{x, y, z\}$: in the same way, z and y are both chosen from $\{y, z\}$, so y is chosen from $\{x, y, z\}$ too; and then so is x.
- $C(\{x, y, z\})$ is not empty, so one of x, y and z is chosen. Whichever it is, x is chosen. □

Profiles

- A group of voters, each with a preference ordering of the options: complete and transitive.
- A *profile*: one ordering for each voter, $(\succsim_1, \ldots, \succsim_n)$.
- For example, Condorcet's profile:[1]

Voters	Ranking
Voter 1	$x \succ y \succ z$
Voter 2	$y \succ z \succ x$
Voter 3	$z \succ x \succ y$

[1] Condorcet (1785).

Social Choice Rules and Social Ordering Functions

Social choice rule

For each profile, a social choice function: society's choice function, saying what may be chosen from each menu.

Social ordering function

For each profile, a social ordering $\succsim$: complete and transitive.

- Arrow's theorem is about social ordering functions; Sen's result is about social choice rules.

Arrow's Conditions

Unrestricted Domain

Every profile is allowed: voters may hold any orderings at all.

Pareto

If every voter prefers x to y, then $x \succ y$.

Independence of Irrelevant Alternatives (IIA)

Take any two profiles, and any two options x and y. If every voter ranks x against y the same way in both profiles, then society ranks x against y the same way in both.

Decisiveness and Dictators

Decisiveness

A voter i is *decisive* over a pair $\{x, y\}$ when, at every profile, whichever of the two they prefer, society prefers too: if $x \succ_i y$, then $x \succ y$; and if $y \succ_i x$, then $y \succ x$.

Dictator

A voter who is decisive over every pair.

Non-dictatorship

No voter is a dictator.

Arrow's Theorem

Arrow's Theorem

With finitely many voters and at least three options, no social ordering function satisfies Unrestricted Domain, Pareto, IIA and Non-dictatorship.[1]

[1] Arrow (1951).

IIA in Action

	Voter 1	Voter 2	Voter 3
P	$x \succ z \succ y$	$y \succ x \succ z$	$x \succ y \succ z$

Two profiles, P and Q.

IIA in Action

	Voter 1	Voter 2	Voter 3
P	$x \succ z \succ y$	$y \succ x \succ z$	$x \succ y \succ z$
Q	$z \succ x \succ y$	$y \succ z \succ x$	$x \succ y \succ z$

In Q, voters 1 and 2 have moved z.

IIA in Action

	Voter 1	Voter 2	Voter 3
P	$x \succ z \succ y$	$y \succ x \succ z$	$x \succ y \succ z$
Q	$z \succ x \succ y$	$y \succ z \succ x$	$x \succ y \succ z$

But every voter ranks x against y the same way in P and Q: voters 1 and 3 prefer x, voter 2 prefers y.

IIA in Action

	Voter 1	Voter 2	Voter 3
P	$x \succ z \succ y$	$y \succ x \succ z$	$x \succ y \succ z$
Q	$z \succ x \succ y$	$y \succ z \succ x$	$x \succ y \succ z$

So society ranks x against y the same way in P and Q (IIA).
Where z is doesn't matter.

What IIA Lets Us Do

- We can change a profile however we like. As long as nobody changes how they rank x against y, society's ranking of x against y stays the same.
- So we can take what society says in one profile, and carry it across to another.
- The proof does this again and again. Each time, we name the two profiles we are comparing.

The Proof: Plan

1. The Extreme Option Lemma: if every voter puts b at the very top or the very bottom, so does society.
2. Moving b from bottom to top one voter at a time, we find a pivotal voter for b.[1]
3. A pivotal voter for b is decisive over every pair not involving b.
4. Every option has a pivotal voter, and they are all the same voter.
5. So this pivotal voter is a dictator.

[1] Steps 1–3 follow Geanakoplos (2005: 212–213).

The Extreme Option Lemma

The Extreme Option Lemma

Suppose a social ordering function satisfies Unrestricted Domain, Pareto and IIA, and every voter ranks b strictly above all other options, or strictly below all other options. Then society ranks b strictly above all other options, or strictly below all other options.

- Even if some voters put b at the top and the rest put b at the bottom.

The Extreme Option Lemma: An Example

	Voter 1	Voter 2	Voter 3
P	$b \succ a \succ c$	$a \succ c \succ b$	$c \succ a \succ b$

In P, every voter puts b at the very top or the very bottom.
Suppose society has $a \succsim b$ and $b \succsim c$.

The Extreme Option Lemma: An Example

	Voter 1	Voter 2	Voter 3
P	$b \succ a \succ c$	$a \succ c \succ b$	$c \succ a \succ b$
P'	$b \succ a \succ c$	$a \succ c \succ b$	$c \succ a \succ b$

P': every voter moves c to just above a.

The Extreme Option Lemma: An Example

	Voter 1	Voter 2	Voter 3
P	$b \succ a \succ c$	$a \succ c \succ b$	$c \succ a \succ b$
P'	$b \succ a \succ \boldsymbol{c}$	$a \succ c \succ b$	$c \succ a \succ b$

P': every voter moves c to just above a.

The Extreme Option Lemma: An Example

	Voter 1	Voter 2	Voter 3
P	$b \succ a \succ c$	$a \succ c \succ b$	$c \succ a \succ b$
P'	$b \succ \boldsymbol{c} \succ a$	$a \succ c \succ b$	$c \succ a \succ b$

P': every voter moves c to just above a.

The Extreme Option Lemma: An Example

	Voter 1	Voter 2	Voter 3
P	$b \succ a \succ c$	$a \succ c \succ b$	$c \succ a \succ b$
P'	$b \succ c \succ a$	$a \succ \boldsymbol{c} \succ b$	$c \succ a \succ b$

P': every voter moves c to just above a.

The Extreme Option Lemma: An Example

	Voter 1	Voter 2	Voter 3
P	$b \succ a \succ c$	$a \succ c \succ b$	$c \succ a \succ b$
P'	$b \succ c \succ a$	$\boldsymbol{c} \succ a \succ b$	$c \succ a \succ b$

P': every voter moves c to just above a.

The Extreme Option Lemma: An Example

	Voter 1	Voter 2	Voter 3
P	$b \succ a \succ c$	$a \succ c \succ b$	$c \succ a \succ b$
P'	$b \succ c \succ a$	$c \succ a \succ b$	$c \succ a \succ b$

IIA: $a \succsim b$ and $b \succsim c$ in P' too. So $a \succsim c$.

The Extreme Option Lemma: An Example

	Voter 1	Voter 2	Voter 3
P	$b \succ a \succ c$	$a \succ c \succ b$	$c \succ a \succ b$
P'	$b \succ c \succ a$	$c \succ a \succ b$	$c \succ a \succ b$

Pareto: $c \succ a$ in P'. Contradiction.

Finding a Pivotal Voter

	Voter 1	Voter 2	Voter 3	Society
P_0	$c \succ a \succ b$	$a \succ c \succ b$	$c \succ a \succ b$	
P_1	$b \succ c \succ a$	$a \succ c \succ b$	$c \succ a \succ b$	
P_2	$b \succ c \succ a$	$b \succ a \succ c$	$c \succ a \succ b$	
P_3	$b \succ c \succ a$	$b \succ a \succ c$	$b \succ c \succ a$	

In P_0, every voter puts b at the bottom.
Then they move b to the top one at a time: voter 1, then 2, then 3.

Finding a Pivotal Voter

	Voter 1	Voter 2	Voter 3	Society
P_0	$c \succ a \succ b$	$a \succ c \succ b$	$c \succ a \succ b$	b at the bottom
P_1	$b \succ c \succ a$	$a \succ c \succ b$	$c \succ a \succ b$	
P_2	$b \succ c \succ a$	$b \succ a \succ c$	$c \succ a \succ b$	
P_3	$b \succ c \succ a$	$b \succ a \succ c$	$b \succ c \succ a$	b at the top

In P_0 every voter has b at the bottom, so society does too;
in P_3 every voter has b at the top, so society does too (Pareto).

Finding a Pivotal Voter

	Voter 1	Voter 2	Voter 3	Society
P_0	$c \succ a \succ b$	$a \succ c \succ b$	$c \succ a \succ b$	b at the bottom
P_1	$b \succ c \succ a$	$a \succ c \succ b$	$c \succ a \succ b$	
P_2	$b \succ c \succ a$	$b \succ a \succ c$	$c \succ a \succ b$	
P_3	$b \succ c \succ a$	$b \succ a \succ c$	$b \succ c \succ a$	b at the top

In between, society has b strictly at the top or bottom (Extreme Option Lemma), so at some move its b jumps all the way up.

Finding a Pivotal Voter

	Voter 1	Voter 2	Voter 3	Society
P_0	$c \succ a \succ b$	$a \succ c \succ b$	$c \succ a \succ b$	b at the bottom
b-BOTTOM	$b \succ c \succ a$	$a \succ c \succ b$	$c \succ a \succ b$	b at the bottom
b-TOP	$b \succ c \succ a$	$b \succ a \succ c$	$c \succ a \succ b$	b at the top
P_3	$b \succ c \succ a$	$b \succ a \succ c$	$b \succ c \succ a$	b at the top

Say it jumps when voter 2 moves: voter 2 is a *pivotal voter* for b.
Call these two profiles b-BOTTOM and b-TOP.

Finding a Pivotal Voter

	Voter 1	Voter 2	Voter 3	Society
P_0	$c \succ a \succ b$	$a \succ c \succ b$	$c \succ a \succ b$	b at the bottom
b-BOTTOM	$b \succ c \succ a$	$a \succ c \succ b$	$c \succ a \succ b$	b at the bottom
b-TOP	$b \succ c \succ a$	$b \succ a \succ c$	$c \succ a \succ b$	b at the top
P_3	$b \succ c \succ a$	$b \succ a \succ c$	$b \succ c \succ a$	b at the top

For all we know so far, with the voters moving in a different order, a different voter might turn out to be pivotal.

Voter 2 Is Decisive Over $\{a, c\}$

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$

- ORIGINAL: any profile in which voter 2 ranks a above c. Here voters 1 and 3 rank c above a.
- We show: in ORIGINAL, society ranks a above c.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$

Build b-MIDDLE from ORIGINAL by moving b only.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$

Build b-MIDDLE from ORIGINAL by moving b only.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$c \succ \mathbf{b} \succ a$	$a \succ c \succ b$	$b \succ c \succ a$

Voter 1 (before voter 2): b to the top.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$\boldsymbol{b} \succ c \succ a$	$a \succ c \succ b$	$b \succ c \succ a$

Voter 1 (before voter 2): b to the top.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$b \succ c \succ a$	$a \succ c \succ b$	$b \succ c \succ a$

Voter 1 (before voter 2): b to the top.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$b \succ c \succ a$	$a \succ c \succ \mathbf{b}$	$b \succ c \succ a$

Voter 2: b between a and c.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$b \succ c \succ a$	$a \succ \mathbf{b} \succ c$	$b \succ c \succ a$

Voter 2: b between a and c.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$b \succ c \succ a$

Voter 2: b between a and c.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$\boldsymbol{b} \succ c \succ a$

Voter 3 (after voter 2): b to the bottom.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ \boldsymbol{b}$

Voter 3 (after voter 2): b to the bottom.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$

Voter 3 (after voter 2): b to the bottom.

Building b-MIDDLE

	Voter 1	Voter 2	Voter 3
ORIGINAL	$c \succ b \succ a$	$a \succ c \succ b$	$b \succ c \succ a$
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$

Only b moved, so society ranks a against c the same way in both (IIA).

Society Ranks a Above b in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	

Start from b-MIDDLE.

Society Ranks a Above b in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	
	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	

Voter 2 moves b to the bottom.

Society Ranks a Above b in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	
	$b \succ c \succ a$	$a \succ \boldsymbol{b} \succ c$	$c \succ a \succ b$	

Voter 2 moves b to the bottom.

Society Ranks a Above b in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	
	$b \succ c \succ a$	$a \succ c \succ \boldsymbol{b}$	$c \succ a \succ b$	

Voter 2 moves b to the bottom.

Society Ranks a Above b in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	
b-BOTTOM	$b \succ c \succ a$	$a \succ c \succ b$	$c \succ a \succ b$	b at the bottom

This is b-BOTTOM: society puts b at the bottom.

Society Ranks a Above b in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$\boldsymbol{b} \succ c \succ \boldsymbol{a}$	$\boldsymbol{a} \succ \boldsymbol{b} \succ c$	$c \succ \boldsymbol{a} \succ \boldsymbol{b}$	
b-BOTTOM	$\boldsymbol{b} \succ c \succ \boldsymbol{a}$	$\boldsymbol{a} \succ c \succ \boldsymbol{b}$	$c \succ \boldsymbol{a} \succ \boldsymbol{b}$	$a \succ b$

Every voter ranks a against b the same way in both.

Society Ranks a Above b in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$\boldsymbol{b} \succ c \succ \boldsymbol{a}$	$\boldsymbol{a} \succ \boldsymbol{b} \succ c$	$c \succ \boldsymbol{a} \succ \boldsymbol{b}$	$a \succ b$
b-BOTTOM	$\boldsymbol{b} \succ c \succ \boldsymbol{a}$	$\boldsymbol{a} \succ c \succ \boldsymbol{b}$	$c \succ \boldsymbol{a} \succ \boldsymbol{b}$	$a \succ b$

So $a \succ b$ in b-MIDDLE (IIA).

Society Ranks b Above c in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	

Start from b-MIDDLE.

Society Ranks b Above c in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	
	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	

Voter 2 moves b to the top.

Society Ranks b Above c in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	
	$b \succ c \succ a$	$a \succ \mathbf{b} \succ c$	$c \succ a \succ b$	

Voter 2 moves b to the top.

Society Ranks b Above c in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	
	$b \succ c \succ a$	$\boldsymbol{b} \succ a \succ c$	$c \succ a \succ b$	

Voter 2 moves b to the top.

Society Ranks b Above c in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$b \succ c \succ a$	$a \succ b \succ c$	$c \succ a \succ b$	
b-TOP	$b \succ c \succ a$	$b \succ a \succ c$	$c \succ a \succ b$	b at the top

This is b-TOP: society puts b at the top.

Society Ranks b Above c in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$\boldsymbol{b} \succ \boldsymbol{c} \succ a$	$a \succ \boldsymbol{b} \succ \boldsymbol{c}$	$\boldsymbol{c} \succ a \succ \boldsymbol{b}$	
b-TOP	$\boldsymbol{b} \succ \boldsymbol{c} \succ a$	$\boldsymbol{b} \succ a \succ \boldsymbol{c}$	$\boldsymbol{c} \succ a \succ \boldsymbol{b}$	$b \succ c$

Every voter ranks b against c the same way in both.

Society Ranks b Above c in b-MIDDLE

	Voter 1	Voter 2	Voter 3	Society
b-MIDDLE	$\boldsymbol{b} \succ \boldsymbol{c} \succ a$	$a \succ \boldsymbol{b} \succ \boldsymbol{c}$	$\boldsymbol{c} \succ a \succ \boldsymbol{b}$	$b \succ c$
b-TOP	$\boldsymbol{b} \succ \boldsymbol{c} \succ a$	$\boldsymbol{b} \succ a \succ \boldsymbol{c}$	$\boldsymbol{c} \succ a \succ \boldsymbol{b}$	$b \succ c$

So $b \succ c$ in b-MIDDLE (IIA).

Society Ranks a Above c in ORIGINAL

- In b-MIDDLE: $a \succ b$ and $b \succ c$, so $a \succ c$.
- So $a \succ c$ in ORIGINAL too (IIA): society goes with voter 2.
- Any ORIGINAL works, and with a and c swapped: voter 2 is decisive over $\{a, c\}$. In general, a pivotal voter for b is decisive over every pair not involving b.

From Here to a Dictator

Pivotal voter for	Decisive over	Give them
c	$\{a, b\}$	
a	$\{b, c\}$	
b	$\{a, c\}$	

Every option has a pivotal voter, decisive over the pairs not involving it.

From Here to a Dictator

Pivotal voter for	Decisive over	Give them
c	$\{a, b\}$	$a \succ b$
a	$\{b, c\}$	
b	$\{a, c\}$	

Suppose the pivotal voters have these preferences.

From Here to a Dictator

Pivotal voter for	Decisive over	Give them
c	$\{a, b\}$	$a \succ b$
a	$\{b, c\}$	$b \succ c$
b	$\{a, c\}$	

Suppose the pivotal voters have these preferences.

From Here to a Dictator

Pivotal voter for	Decisive over	Give them
c	$\{a, b\}$	$a \succ b$
a	$\{b, c\}$	$b \succ c$
b	$\{a, c\}$	$c \succ a$

Suppose the pivotal voters have these preferences.

From Here to a Dictator

Pivotal voter for	Decisive over	Give them
c	$\{a, b\}$	$a \succ b$
a	$\{b, c\}$	$b \succ c$
b	$\{a, c\}$	$c \succ a$

Then society has $a \succ b$, $b \succ c$ and $c \succ a$: not an ordering.

From Here to a Dictator

Pivotal voter for	Decisive over	Give them
c	$\{a, b\}$	$a \succ b$
a	$\{b, c\}$	$b \succ c$
b	$\{a, c\}$	$c \succ a$

With at least two different pivotal voters here, these preferences fit into orderings: nothing stops them having them.

From Here to a Dictator

Pivotal voter for	Decisive over	Give them
c	$\{a, b\}$	$a \succ b$
a	$\{b, c\}$	$b \succ c$
b	$\{a, c\}$	$c \succ a$

With just one, they would form a cycle, which no voter can hold.
So there is just one pivotal voter, decisive over every pair: a dictator.

The Conditions

Pareto

If everyone strictly prefers x to y, then y is not chosen from any menu containing x.

Decisiveness

A person is *decisive* over a pair $\{x, y\}$ when, whichever of the two they prefer, the other is not chosen from any menu containing both.

Minimal Liberalism

There are at least two people who are each decisive over at least one pair of social states.

Sen's Result

The liberal paradox

If any preferences at all are allowed (unrestricted domain), no social choice rule satisfies both Pareto and Minimal Liberalism.[1]

- Minimal Liberalism gives two people: person 1, decisive over one pair, and person 2, decisive over a pair (perhaps the same one).
- The two pairs are the same, or they share one option, or they share none. In each case, we pick preferences so that every option on some menu is excluded.

[1] Sen (1970: 154).

Sen's Proof: The Same Pair

x

y

Person 1, decisive over $\{x, y\}$:
$x \succ y$

Person 2, decisive over $\{x, y\}$:
$y \succ x$

Sen's Proof: The Same Pair

x

~~y~~

Person 1, decisive over $\{x, y\}$:
$\boldsymbol{x \succ y}$

Person 2, decisive over $\{x, y\}$:
$y \succ x$

Person 1 is decisive over $\{x, y\}$: y is out.

Sen's Proof: The Same Pair

~~x~~

~~y~~

Person 1, decisive over $\{x, y\}$:
$x \succ y$

Person 2, decisive over $\{x, y\}$:
$\boldsymbol{y} \succ \boldsymbol{x}$

Person 2 is decisive over $\{x, y\}$: x is out.

Sen's Proof: The Same Pair

~~x~~

~~y~~

Person 1, decisive over $\{x, y\}$:
$x \succ y$

Person 2, decisive over $\{x, y\}$:
$y \succ x$

Nothing can be chosen from $\{x, y\}$.

Sen's Proof: One Shared Option

x

y

z

Person 1, decisive over $\{x, z\}$:
$z \succ x \succ y$

Person 2, decisive over $\{y, z\}$:
$x \succ y \succ z$

Everyone else:
$x \succ y$

Sen's Proof: One Shared Option

~~x~~

y

z

Person 1, decisive over $\{x, z\}$:
$\boldsymbol{z} \succ \boldsymbol{x} \succ y$

Person 2, decisive over $\{y, z\}$:
$x \succ y \succ z$

Everyone else:
$x \succ y$

Person 1 is decisive over $\{x, z\}$: x is out.

Sen's Proof: One Shared Option

~~x~~

y

~~z~~

Person 1, decisive over $\{x, z\}$:
$z \succ x \succ y$

Person 2, decisive over $\{y, z\}$:
$x \succ \boldsymbol{y} \succ \boldsymbol{z}$

Everyone else:
$x \succ y$

Person 2 is decisive over $\{y, z\}$: z is out.

Sen's Proof: One Shared Option

~~x~~

~~y~~

~~z~~

Person 1, decisive over $\{x, z\}$:
$z \succ \boldsymbol{x} \succ \boldsymbol{y}$

Person 2, decisive over $\{y, z\}$:
$\boldsymbol{x} \succ \boldsymbol{y} \succ z$

Everyone else:
$\boldsymbol{x} \succ \boldsymbol{y}$

Pareto: everyone prefers x to y, so y is out.

Sen's Proof: One Shared Option

~~x~~

~~y~~

~~z~~

Person 1, decisive over $\{x, z\}$:
$z \succ x \succ y$

Person 2, decisive over $\{y, z\}$:
$x \succ y \succ z$

Everyone else:
$x \succ y$

Nothing can be chosen from $\{x, y, z\}$.

Sen's Proof: No Shared Option

x

y

z

w

Person 1, decisive over $\{x, y\}$:
$w \succ x \succ y \succ z$

Person 2, decisive over $\{z, w\}$:
$y \succ z \succ w \succ x$

Everyone else:
$w \succ x, \quad y \succ z$

Sen's Proof: No Shared Option

x

~~y~~

z

w

Person 1, decisive over $\{x, y\}$:
$w \succ \boldsymbol{x} \succ \boldsymbol{y} \succ z$

Person 2, decisive over $\{z, w\}$:
$y \succ z \succ w \succ x$

Everyone else:
$w \succ x, \quad y \succ z$

Person 1 is decisive over $\{x, y\}$: y is out.

Sen's Proof: No Shared Option

x

~~y~~

z

~~w~~

Person 1, decisive over $\{x, y\}$:
$w \succ x \succ y \succ z$

Person 2, decisive over $\{z, w\}$:
$y \succ \boldsymbol{z} \succ \boldsymbol{w} \succ x$

Everyone else:
$w \succ x, \quad y \succ z$

Person 2 is decisive over $\{z, w\}$: w is out.

Sen's Proof: No Shared Option

~~x~~

~~y~~

z

~~w~~

Person 1, decisive over $\{x, y\}$:
$\boldsymbol{w} \succ \boldsymbol{x} \succ y \succ z$

Person 2, decisive over $\{z, w\}$:
$y \succ z \succ \boldsymbol{w} \succ \boldsymbol{x}$

Everyone else:
$\boldsymbol{w} \succ \boldsymbol{x}, \quad y \succ z$

Pareto: everyone prefers w to x, so x is out.

Sen's Proof: No Shared Option

~~x~~

~~y~~

~~z~~

~~w~~

Person 1, decisive over $\{x, y\}$:
$w \succ x \succ \boldsymbol{y} \succ \boldsymbol{z}$

Person 2, decisive over $\{z, w\}$:
$\boldsymbol{y} \succ \boldsymbol{z} \succ w \succ x$

Everyone else:
$w \succ x$, $\boldsymbol{y} \succ \boldsymbol{z}$

Pareto: everyone prefers y to z, so z is out.

Sen's Proof: No Shared Option

~~x~~

~~y~~

~~z~~

~~w~~

Person 1, decisive over $\{x, y\}$:
$w \succ x \succ y \succ z$

Person 2, decisive over $\{z, w\}$:
$y \succ z \succ w \succ x$

Everyone else:
$w \succ x, \quad y \succ z$

Nothing can be chosen from $\{x, y, z, w\}$.

Gibbard's Paradox: The Point

- Sen understands rights as individual decisiveness.
- Take a case where people clearly do have rights, and read them as decisiveness. With unrestricted domain, the rights are inconsistent.[1]
- In Sen's paradox, several plausible conditions clash, and it isn't clear which must go. Here it is clear: liberalism isn't inconsistent, so this is the wrong way to understand rights.

[1] Gibbard (1974: 389).

Gibbard's Paradox: The Walls

- Nick and Carl each have the right to paint their own walls white (W) or yellow (Y).
- WW: both white. WY: Nick's white, Carl's yellow. And so on.
- Rights as decisiveness: of two states that differ only in Nick's walls, the one Nick prefers excludes the other. Likewise for Carl.

The Walls: Every State Is Excluded

WW WY

YW YY

Nick:

$YW \succ WW$

$WY \succ YY$

Carl:

$WW \succ WY$

$YY \succ YW$

The Walls: Every State Is Excluded

~~WW~~ WY

YW YY

Nick:
YW $\succ$ **WW**
WY $\succ$ YY

Carl:
WW $\succ$ WY
YY $\succ$ YW

Nick: YW excludes WW.

The Walls: Every State Is Excluded

~~WW~~ ~~WY~~

YW YY

Nick:
YW $\succ$ WW
WY $\succ$ YY

Carl:
WW $\succ$ **WY**
YY $\succ$ YW

Carl: WW excludes WY.

The Walls: Every State Is Excluded

~~WW~~ ~~WY~~

YW ~~YY~~

Nick:
YW $\succ$ WW
WY $\succ$ **YY**

Carl:
WW $\succ$ WY
YY $\succ$ YW

Nick: WY excludes YY.

The Walls: Every State Is Excluded

~~WW~~ ~~WY~~

~~YW~~ ~~YY~~

Nick:
YW $\succ$ WW
WY $\succ$ YY

Carl:
WW $\succ$ WY
YY $\succ$ **YW**

Carl: YY excludes YW.

The Walls: Every State Is Excluded

~~WW~~ ~~WY~~

~~YW~~ ~~YY~~

Nick:
$YW \succ WW$
$WY \succ YY$

Carl:
$WW \succ WY$
$YY \succ YW$

Every state is excluded, by rights alone.

References

Arrow, Kenneth J. (1951). *Social Choice and Individual Values*. New York: Wiley.

Condorcet, Marquis de (1785). *Essai sur l'application de l'analyse à la probabilité des décisions rendues à la pluralité des voix*. Paris: Imprimerie Royale.

Geanakoplos, John (2005). "Three Brief Proofs of Arrow's Impossibility Theorem". In: *Economic Theory* 26.1, pp. 211–215.

Gibbard, Allan (1974). "A Pareto-Consistent Libertarian Claim". In: *Journal of Economic Theory* 7.4, pp. 388–410.

Sen, Amartya (1970). "The Impossibility of a Paretian Liberal". In: *Journal of Political Economy* 78.1, pp. 152–157.

Sen, Amartya (1971). "Choice Functions and Revealed Preference". In: *The Review of Economic Studies* 38.3, pp. 307–317.

Sen, Amartya (1993). "Internal Consistency of Choice". In: *Econometrica* 61.3, pp. 495–521.